Non-Homomorphism Factors and the Concentration Parameter in Graph Homomorphisms

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Abstract

The *non-homomorphism factor* $|G, H|$ is the minimum number of edges that must be removed from G so that the remaining graph admits a homomorphism to H. The *concentration parameter* $\gamma(G, H)$ is the minimum, over all homomorphisms $f\colon G \to H$, of the largest number of edges of G that f sends onto a single edge of H.

For $|G, H|$ we prove $|K_n, K_m| = \binom{n}{2} - t(n, m)$ and $|K_n, H| = \binom{n}{2} - \lfloor n^2/4 \rfloor$ for every triangle-free H with an edge. For a d-regular graph G on n vertices and triangle-free H we prove $|L(G), H| \geq n\lfloor (d-1)^2/4 \rfloor$, with equality when G has a balanced 2-edge-colouring. We prove the quantitative Hedetniemi bound

$$|G \times H, K| \leq \min(2|E(H)|\,|G, K|,\, 2|E(G)|\,|H, K|)$$

and the exact value $|C_p \times C_q, K_2| = 2p$ for odd $p \leq q$. For cubic G we show that $|L(G), L(P)|$, where P is the Petersen graph, is at least the number of leaf blocks of the bridge-block tree of G.

For γ we prove multiplicativity, monotonicity, and the transfer bound $|G, K| \leq \gamma(G, H)\,|H, K|$. We find an exact identity for the loads of a homomorphism to K_3 in terms of the degree sums of its colour classes. It gives exact values of $\gamma(\cdot, K_3)$ for cycles, complete bipartite graphs, the Petersen graph, books, friendship graphs, wheels and complete tripartite graphs. We also show that $\max(|G, H|, |H, G|)$ is not a pseudo-metric.

Every result in this paper is formally verified in the Lean 4 proof assistant with the Mathlib library.

1 Introduction

All graphs are finite, simple and undirected.

Definition 1.1 (Graph homomorphism)**.** A *homomorphism* from G to H is a map $f\colon V(G) \to V(H)$ such that $\{u, v\} \in E(G)$ implies $\{f(u), f(v)\} \in E(H)$. We write $G \to H$ when such a map exists, and $G \leftrightarrow H$ when $G \to H$ and $H \to G$.

A proper k-colouring of G is the same as a homomorphism $G \to K_k$. Two standard facts [3] follow directly.

Proposition 1.2. *If $G \to H$ then $\chi(G) \leq \chi(H)$.*

Proposition 1.3. *Every homomorphism $f\colon G \to H$ satisfies $d_H(f(u), f(v)) \leq d_G(u, v)$ for all $u, v \in V(G)$.*

Proof. A walk of length ℓ from u to v maps to a walk of length ℓ from $f(u)$ to $f(v)$. □

Formal verification. Every theorem, lemma, proposition and corollary in this paper has been proved in Lean 4 [9] using Mathlib [10]. The only exceptions are the two classical results quoted in Remark 2.9. The development is in the `lean/NonHom` directory of the repository. Section 5 describes it and lists the Lean name of each result.

2 Non-Homomorphism Factors

Definition 2.1 (Non-homomorphism factor [1])**.** $|G, H|$ is the minimum number of edges that must be removed from G to obtain a graph G' with $G' \to H$. In particular $|G, H| = 0$ if and only if $G \to H$.

Remark 2.2 (Complexity)**.** For fixed H, deciding $|G, H| \le k$ is Edge Deletion-H-Colouring, studied by Foucaud, Hocquard, Lajou, Mitsou and Pierron [4]. The case $k = 0$ is H-Colouring. So by the Hell–Nešetřil dichotomy [3], computing $|G, H|$ is NP-hard whenever the core of H has more than one edge. For $H = K_2$ the problem is Edge Bipartization, which is NP-complete but fixed-parameter tractable in k.

Proposition 2.3 ([1])**.** *1. If $H_1 \to H_2$ then $|G, H_1| \ge |G, H_2|$ for every G.*

2. If $H_1 \leftrightarrow H_2$ then $|G, H_1| = |G, H_2|$ for every G.

3. If $f\colon G_1 \to G_2$ is edge-injective (distinct edges map to distinct edges) then $|G_1, H| \le |G_2, H|$ for every H.

Proof. (1) If $G' \to H_1$ then $G' \to H_1 \to H_2$. (2) Apply (1) in both directions. (3) Take $S \subseteq E(G_2)$ with $|S| = |G_2, H|$ and $G_2 - S \to H$, and let $S' = \{e \in E(G_1) : f(e) \in S\}$. Edge-injectivity gives $|S'| \le |S|$, and f restricts to a homomorphism $G_1 - S' \to G_2 - S$. □

2.1 Complete graphs and Turán's theorem

Let $t(n, m)$ be the number of edges of the Turán graph $T(n, m)$, the complete m-partite graph on n vertices with parts as equal as possible. By Turán's theorem, every K_{m+1}-free graph on n vertices has at most $t(n, m)$ edges.

Theorem 2.4. *For $m \ge 1$, $|K_n, K_m| = \binom{n}{2} - t(n, m)$. If $n = qm + r$ with $0 \le r < m$, this equals $r\binom{q+1}{2} + (m - r)\binom{q}{2}$. In particular $|K_{n+1}, K_n| = 1$ for all $n \ge 1$.*

Proof. Removing the edges outside $T(n, m)$ leaves an m-colourable graph. Conversely, an m-colourable spanning subgraph of K_n is K_{m+1}-free, so by Turán's theorem it has at most $t(n, m)$ edges. The explicit formula follows by induction on q from $t(n+m, m) = t(n, m) + n(m-1) + \binom{m}{2}$. For $n + 1 = 1 \cdot n + 1$ it gives $\binom{2}{2} = 1$. □

Table 1: $|K_n, K_m|$ for $n = 2, \dots, 7$ and $m = 2, \dots, 6$, from Theorem 2.4.

	K_2	K_3	K_4	K_5	K_6
K_2	0	0	0	0	0
K_3	1	0	0	0	0
K_4	2	1	0	0	0
K_5	4	2	1	0	0
K_6	6	3	2	1	0
K_7	9	5	3	2	1

Theorem 2.5. *For all graphs G and H, $|G, H| \ge \max\big(0,\ |E(G)| - t(|V(G)|, \chi(H))\big)$.*

Proof. If $G' \subseteq G$ and $G' \to H$ then $\chi(G') \le \chi(H)$, so G' is $K_{\chi(H)+1}$-free and has at most $t(|V(G)|, \chi(H))$ edges. □

Triangle-free targets all give the same answer for complete graphs.

Theorem 2.6. *If H is triangle-free and has at least one edge, then $|K_n, H| = \binom{n}{2} - \lfloor n^2/4 \rfloor$.*

Proof. If $G' \to H$ then G' is triangle-free, because a homomorphism maps a triangle onto a triangle. So by Mantel's theorem $|E(G')| \le \lfloor n^2/4 \rfloor$. Conversely $K_n - E(\overline{T(n, 2)}) = T(n, 2) \to K_2 \to H$. □

2.2 Cycles and the Petersen graph

Every cycle is 3-colourable, so $|C_n, K_3| = 0$. For K_2, $|C_n, K_2| = 0$ if n is even and $|C_n, K_2| = 1$ if $n \geq 3$ is odd. Indeed an odd cycle is not bipartite, and deleting one edge leaves a path.

Let P be the Petersen graph, with outer cycle $u_0u_1u_2u_3u_4$, spokes u_iv_i, and inner edges v_iv_{i+2} (indices mod 5). It has 10 vertices, 15 edges, and $\chi(P) = 3$, so $|P, K_m| = 0$ for $m \geq 3$.

Theorem 2.7. $|P, K_2| = 3$.

Proof. P has exactly twelve 5-cycles, and each edge lies on exactly four of them. Two edges therefore lie on at most eight, so after deleting any two edges some 5-cycle survives and the graph is not bipartite. Hence $|P, K_2| \geq 3$. Conversely, deleting u_0u_1, u_3v_3 and v_2v_4 leaves a bipartite graph with sides $\{u_0, u_1, u_3, v_2, v_3, v_4\}$ and $\{u_2, u_4, v_0, v_1\}$. □

Theorem 2.5 only gives $|P, K_2| \geq 15 - t(10, 2) < 0$ here, so the odd-cycle structure, not edge density, is what forces the deletions.

2.3 Line graphs of regular graphs

Theorem 2.8. *Let G be d-regular on n vertices and let H be triangle-free. Then*

$$|L(G), H| \;\geq\; n\left(\binom{d}{2} - \left\lfloor \frac{d^2}{4} \right\rfloor\right) \;=\; n\left\lfloor \frac{(d-1)^2}{4} \right\rfloor.$$

Suppose moreover that H has an edge and that $E(G)$ has a 2-colouring that is balanced*, meaning that at every vertex the two colours occur $\lfloor d/2 \rfloor$ and $\lceil d/2 \rceil$ times. Then equality holds.*

Proof. The d edges at a vertex v form a clique $Q_v \cong K_d$ in $L(G)$. Distinct cliques share no edge of $L(G)$, since two distinct edges of G share at most one vertex. If $L(G) - S \to H$ then $L(G) - S$ is triangle-free. By Mantel's theorem it keeps at most $\lfloor d^2/4 \rfloor$ of the $\binom{d}{2}$ edges of each Q_v. Summing over v gives the lower bound.

For the upper bound, colour the vertices of $L(G)$ by the given edge colouring, and delete the edges of $L(G)$ whose ends have the same colour. What remains maps to $K_2 \to H$. At a vertex with colour counts a and $d - a$ the deleted edges number $\binom{a}{2} + \binom{d-a}{2} = \binom{d}{2} - a(d-a)$. This equals $\binom{d}{2} - \lfloor d^2/4 \rfloor$ when the colouring is balanced. □

Remark 2.9. Balanced 2-edge-colourings exist in three cases. If $4 \mid d$, split a 2-factorization (Petersen's 2-factor theorem [6]) evenly. If G is bipartite, group the d colour classes of a proper edge colouring (Kőnig [7]). If d is odd and G has a perfect matching M, split $G - M$ into $(d-1)/2$ two-factors. Then give colour 1 to M together with $\lfloor (d-1)/4 \rfloor$ of these two-factors.

In particular every bridgeless cubic graph G on n vertices has a perfect matching, by Petersen's theorem [6]. So $|L(G), H| = n$ for every triangle-free H with an edge, and the whole ladder $|L(G), C_5| = |L(G), C_7| = \cdots = |L(G), K_2|$ collapses.

These two classical theorems are not formalized. The Lean proof establishes the cubic case for every cubic graph that has a perfect matching.

Theorem 2.10. *Let C_n^2 ($n \geq 5$) be the circulant graph on $\mathbb{Z}_n$ with jumps 1 and 2. For every triangle-free H, $|C_n^2, H| \geq \lceil n/2 \rceil$.*

Proof. The n triangles $\{i, i+1, i+2\}$ must each lose an edge. An edge $\{i, i+1\}$ lies only in the triangles starting at $i-1$ and at i, and an edge $\{i, i+2\}$ lies only in the triangle starting at i. So each deleted edge destroys at most two of these triangles. □

2.4 Tensor products

The tensor product $G \times H$ has vertex set $V(G) \times V(H)$, with $(g, h) \sim (g', h')$ if and only if $g \sim g'$ and $h \sim h'$.

Theorem 2.11 (Quantitative Hedetniemi bound)**.** *For all graphs G, H and every graph K with at least one vertex,*

$$|G \times H, K| \;\leq\; \min\big(2|E(H)| \cdot |G, K|,\; 2|E(G)| \cdot |H, K|\big).$$

Proof. The projection $G \times H \to G$ sends exactly $2|E(H)|$ edges onto each edge of G, one for each orientation of each edge of H. Now apply Lemma 3.5 to this map. If $G \times H$ has no edges it maps to any graph with a vertex. The second bound is symmetric. □

The hypothesis on K is needed. If K has no vertices and H has no edges, the right-hand side is 0 but the left-hand side is infinite. By Shitov's counterexample to Hedetniemi's conjecture [5], the bound can be far from tight for large targets: the left-hand side can be 0 while neither factor is k-colourable. For $K = K_2$ and two odd cycles it is tight.

Theorem 2.12. *For odd $3 \le p \le q$, $|C_p \times C_q, K_2| = 2p$.*

Proof. The upper bound is Theorem 2.11, since $|C_q, K_2| = 1$.

For the lower bound, let $\varepsilon_i = +1$ for $0 \le i < (q+p)/2$ and $\varepsilon_i = -1$ for $(q+p)/2 \le i < q$, and let $g_i = \sum_{j<i} \varepsilon_j$. Then $g_q = p \equiv 0 \pmod p$. For $a \in \mathbb{Z}_p$, $h \in \mathbb{Z}_q$ and $\sigma \in \{\pm 1\}$, the walk $i \mapsto (a + \sigma g_i,\, h + i)$, $0 \le i \le q$, is a closed walk of odd length q in $C_p \times C_q$. So each of these $2pq$ walks must contain a deleted edge.

Fix a step index i and an edge e. The second coordinate increases by 1 along every step, so it determines which end of e the step leaves from. The first coordinate then determines $\sigma\varepsilon_i$, hence σ, and then a and h. So at most one walk uses e at step i. Charging each walk to the step at which it meets the deletion set S gives $2pq \le q|S|$. □

2.5 Petersen colourings

A *Petersen colouring* of a cubic graph G is a map $E(G) \to E(P)$ that sends the three edges at each vertex onto the three edges at some vertex of P. Since P is triangle-free, every triangle of $L(P)$ is the star of a vertex. So Petersen colourings of G are exactly the homomorphisms $L(G) \to L(P)$, and Jaeger's Petersen colouring conjecture [8] says that $|L(G), L(P)| = 0$ for every bridgeless cubic G.

Theorem 2.13. *Let G be cubic. Let $X_1, \dots, X_t \subseteq V(G)$ be pairwise disjoint sets, each of which is left by exactly one edge of G. Then $|L(G), L(P)| \ge t$. In particular $|L(G), L(P)|$ is at least the number of leaves of the bridge-block tree of G.*

Proof. Suppose $\varphi\colon L(G) - S \to L(P)$. Call $x \in V(G)$ *good* if none of the three edges of $L(G)$ between edges at x lies in S. For a good x, the images of the three edges at x form a triangle of $L(P)$, which is a star.

P has exactly six perfect matchings, and each edge lies in exactly two of them. Fix $X = X_i$, its leaving edge b, and a perfect matching M. Each good x has exactly one incident edge whose image is in M. If every vertex of X is good, counting incidences gives $|X| = 2m + [\varphi(b) \in M]$, where m is the number of edges inside X with image in M. But $3|X| = 2|E(X)| + 1$, so $|X|$ is odd and $\varphi(b) \in M$. This holds for all six matchings, which is impossible.

So each X_i contains a bad vertex x_i, together with an edge $\{e_i, f_i\} \in S$ of $L(G)$ where e_i, f_i are distinct edges at x_i. Two distinct edges of G share at most one vertex, so these t elements of S are distinct. The vertex sets of the leaf blocks of the bridge-block tree satisfy the hypothesis, each being left only by its bridge. □

3 The Concentration Parameter

Definition 3.1 (Concentration parameter)**.** For a homomorphism $f\colon G \to H$ and $e \in E(H)$, the *load* $\mu(f, e)$ is the number of edges of G mapped onto e. Let $\mu(f) = \max_e \mu(f, e)$ and

$$\gamma(G, H) = \min_{f\colon G \to H} \mu(f),$$

with $\gamma(G, H) = +\infty$ when $G \not\to H$.

A related per-homomorphism quantity M^σ appears in [2].

Proposition 3.2 (Pigeonhole bound)**.** *If $G \to H$ then $|E(G)| \le |E(H)| \cdot \gamma(G, H)$, so $\gamma(G, H) \ge \lceil |E(G)|/|E(H)| \rceil$.*

Proof. The loads of f sum to $|E(G)|$. □

Lemma 3.3. *If $|E(G)| \ge 1$ and $G \to H$ then $\gamma(G, H) \ge 1$. Moreover $\gamma(G, G) = 1$.*

Lemma 3.4 (Multiplicativity)**.** *If $G \to H \to K$ then $\gamma(G, K) \le \gamma(G, H) \cdot \gamma(H, K)$.*

Proof. Take optimal $f\colon G \to H$ and $g\colon H \to K$. An edge e of K has at most $\mu(g)$ preimage edges in H, and each of these has at most $\mu(f)$ preimages in G. So $\mu(g \circ f) \le \mu(f)\mu(g)$. □

Lemma 3.5 (Transfer bound)**.** *If $G \to H$ and $|E(G)| \geq 1$, then $|G,K| \leq \gamma(G,H) \cdot |H,K|$ for every K. More generally $|G,K| \leq \mu(f)\,|H,K|$ for every homomorphism $f\colon G \to H$.*

Proof. Let $H - S \to K$ with $|S| = |H,K|$, and let T be the set of edges of G that f maps into S. Then $|T| \leq \mu(f)|S|$, and f restricts to $G - T \to H - S \to K$. □

Corollary 3.6 (Two-sided bound)**.** *If $G \to H$ and $K \to H$, and both have an edge, then* $\max(|G,K|, |K,G|) \leq \max\big(\gamma(G,H)|H,K|,\ \gamma(K,H)|H,G|\big)$.

Lemma 3.7 (Monotonicity in the target)**.** *If H_1 embeds injectively in H_2 (for instance $H_1 \subseteq H_2$), then $\gamma(G,H_2) \leq \gamma(G,H_1)$.*

Proof. Composing with an injective homomorphism does not change any load. □

Proposition 3.8. *For $2 \leq n \leq m$, $\gamma(K_n,K_m) = 1$. For $n > m$, $\gamma(K_n,K_m) = +\infty$.*

3.1 An exact identity for homomorphisms to K_3

For $f\colon G \to K_3$ with colour classes A_0, A_1, A_2, let $D_i = \sum_{v \in A_i} \deg v$ and write $L_{ij} = \mu(f, \{i,j\})$.

Lemma 3.9. *For every graph G, every $f\colon G \to K_3$ and distinct colours a,b,c,*

$$2L_{ab} = D_a + D_b - D_c, \qquad \text{hence} \qquad 2\mu(f) = \max(D_0 + D_1 - D_2,\ D_0 + D_2 - D_1,\ D_1 + D_2 - D_0).$$

Proof. Each edge at a vertex of A_a goes to A_b or to A_c, so $D_a = L_{ab} + L_{ac}$. Solve the three equations. □

Theorem 3.10 (d-regularity formula)**.** *Let G be d-regular on n vertices with $d \geq 1$, and let $f\colon G \to K_3$ have colour classes of sizes a,b,c. Then*

$$2\mu(f) = d \cdot \max(a+b-c,\ a+c-b,\ b+c-a).$$

The triple $(a+b-c, a+c-b, b+c-a)$ is non-negative, sums to n, and has constant parity. Hence, if M is the least maximum of such a triple, then $\gamma(G,K_3) \geq \frac{d}{2}M$. Equality holds as soon as some proper 3-colouring attains M.

Proof. Put $D_i = d|A_i|$ in Lemma 3.9. Non-negativity follows from $d\,(a+b-c) = 2L_{ab} \geq 0$, and the entries differ by even numbers. □

Theorem 3.11. *For all $n \geq 3$,*

$$\gamma(C_n,K_3) = \begin{cases} \lceil n/3 \rceil & \text{if } n \not\equiv 2 \pmod 3, \\ \lceil n/3 \rceil + 1 & \text{if } n \equiv 2 \pmod 3. \end{cases}$$

In particular $\gamma(C_n,K_3)$ attains the pigeonhole bound if and only if $n \not\equiv 2 \pmod 3$.

Proof. Apply Theorem 3.10 with $d = 2$ and write $n = 3q + r$. A same-parity triple summing to n has maximum at least $\lceil n/3 \rceil$. For $r = 2$ the only triple with maximum $q+1$ is $(q+1, q+1, q)$, which has mixed parity, so the maximum is at least $q+2$.

To see that these bounds are attained, colour vertex $i < 3q$ by i mod 3. If $r = 1$, give the last vertex colour 1. If $r = 2$, give the last two vertices colours 0 and 1. The class sizes are (q,q,q), $(q,q+1,q)$ and $(q+1,q+1,q)$, with maxima q, $q+1$ and $q+2$. □

Theorem 3.12. *$|K_{n,n},K_3| = 0$ and $\gamma(K_{n,n},K_3) = n\lceil n/2 \rceil$.*

Proof. Let $f\colon K_{n,n} \to K_3$. The colour sets of the two sides are disjoint, so one side is monochromatic in a colour that the other side does not use. That colour class has exactly n vertices. With $d = n$ and class sizes $(x, n-x, n)$, Theorem 3.10 gives $\mu(f) = n\max(x, n-x) \geq n\lceil n/2 \rceil$. Splitting one side as evenly as possible attains this. □

For $n = 1,2,3,4$ this gives $1,2,6,8$.

Theorem 3.13. *$\gamma(P,K_3) = 6$ and $\gamma(P,K_5) = 2$.*

Proof. For K_3 take $d = 3$ and $n = 10$. Same-parity triples summing to 10 are all even, so their maximum is at least 4, and $\gamma(P,K_3) \geq 6$. Equality holds because P has a proper 3-colouring with class sizes $(3,4,3)$, whose triple is $(4,2,4)$. For K_5, the pigeonhole bound $\lceil 15/10 \rceil = 2$ is attained by an explicit 5-colouring. □

So $\gamma(P,K_3) = 6$ exceeds the pigeonhole bound 5 because of parity. In contrast $\gamma(P,K_5)$ attains it.

3.2 Irregular source graphs

The *book* B_k is k triangles sharing one edge (the spine). The *friendship graph* F_k is k triangles sharing one vertex. The *wheel* W_n is a vertex (the hub) joined to every vertex of C_{n-1}.

Theorem 3.14. *1. For $k \geq 1$: $\gamma(B_k, K_3) = k$ and $|B_k, K_2| = 1$.*

2. For $k \geq 0$: $\gamma(F_k, K_3) = k$ and $|F_k, K_2| = k$.

3. For odd $n \geq 5$: $\gamma(W_n, K_3) = n - 1$.

4. For $p, q, r \geq 1$: $\gamma(K_{p,q,r}, K_3) = \max(pq, pr, qr)$.

Proof. All four use Lemma 3.9.

(1) The two spine vertices get distinct colours a, b, and every page vertex gets the third colour c. So $D_a = D_b = k + 1$ and $D_c = 2k$, giving $L_{ab} = 1$ and $L_{ac} = L_{bc} = k$. Deleting the spine leaves a bipartite graph, and B_k contains a triangle, so $|B_k, K_2| = 1$.

(2) The hub gets some colour c. Each triangle puts one vertex in each of the other two classes, so $D_0 = D_1 = D_2 = 2k$ and every load equals k, for every homomorphism. The k triangles are edge-disjoint, so $|F_k, K_2| \geq k$. Deleting the k edges opposite the hub leaves a star.

(3) The hub, of colour c, is alone in its class, so $D_c = n - 1$. The total degree is $4(n - 1)$, so $D_a + D_b = 3(n - 1)$ and $L_{ab} = n - 1$. Alternating two colours on the even rim attains this.

(4) If all parts are non-empty, each part is monochromatic and the three parts get distinct colours. Otherwise some vertex would see three colours among its neighbours. The class of part i has $D = s_i(N - s_i)$, where $N = p+q+r$, so $2L = s_i(N - s_i) + s_j(N - s_j) - s_k(N - s_k) = 2s_i s_j$. □

In particular $\gamma(F_k, K_3)$ attains the pigeonhole bound k. On the other hand $\gamma(W_n, K_3) = n - 1$ exceeds it for $n \geq 5$, and so do $\gamma(K_{1,2,3}, K_3) = 6 > 4$ and $\gamma(K_{2,3,4}, K_3) = 12 > 9$. The book graphs separate the two parameters: $|B_k, K_2| = 1$ for every k, while $\gamma(B_k, K_3) = k$ grows without bound.

3.3 Cycles into odd cycles

Theorem 3.15. *Let $m \geq 3$ be odd and $n \geq 3$. Then $C_n \to C_m$ if and only if n is even or $n \geq m$.*

Proof. An even cycle maps onto an edge. If $n \geq m$ are both odd, wind once around C_m and then oscillate on an edge for the remaining even number of steps. Conversely, let $n < m$ with n odd and $f : C_n \to C_m$. Then f misses some vertex w, and $C_m - w$ is a path. So C_n would be bipartite, a contradiction. □

When $5 \nmid n$ the pigeonhole bound $\lceil n/5 \rceil$ can still be attained. For example $\gamma(C_6, C_5) = 2$, via $0, 1, 2, 3, 2, 1$. Similarly $\gamma(C_8, C_7) = \gamma(C_{10}, C_7) = \gamma(C_{12}, C_7) = 2$ and $\gamma(C_{15}, C_7) = 3$ all equal $\lceil n/7 \rceil$.

4 The Symmetric Combination $m(G, H)$

Let $m(G, H) = \max(|G, H|, |H, G|)$. It is non-negative and symmetric, and $m(G, G) = 0$.

Proposition 4.1. *$m(G, H) = 0$ if and only if $G \leftrightarrow H$.*

Theorem 4.2. *m does not satisfy the triangle inequality, and so is not a pseudo-metric.*

Proof. By Theorem 2.4, and since $K_n \to K_m$ for $n \leq m$,

$$m(K_7, K_3) = 5, \qquad m(K_7, K_5) = 2, \qquad m(K_5, K_3) = 2,$$

and $5 > 2 + 2$. □

Remark 4.3. Khoshkha [1] claims that m is a pseudo-metric. Theorem 4.2 shows that this is false under Definition 2.1.

5 Formal Verification

The Lean 4 development in `lean/NonHom` uses Lean and Mathlib version 4.35.0. It consists of 17 files and about 3 300 lines. It defines $|G, H|$ as the infimum, in $\mathbb{N} \cup \{\infty\}$, of $|S|$ over finite edge sets S such that $G - S \to H$. It defines $\gamma(G, H)$ as the infimum of $\mu(f)$ over all homomorphisms, which is ∞ when there are none.

There is no `sorry` and no `native_decide`. Every theorem depends only on Lean's standard axioms (`propext`, `Classical.choice`, `Quot.sound`). Finite checks, such as the facts about the 5-cycles and perfect matchings of P and the explicit colourings, are done with `decide +kernel`, which the Lean kernel checks. Turán's theorem and Mantel's theorem are taken from Mathlib. Table 2 lists the Lean name of each result.

Table 2: Lean names of the results.

Result	Lean theorem
Props. 1.2, 1.3	`chromaticNumber_le_of_hom`, `edist_map_le`
Prop. 2.3	`nhf_le_of_hom_target`, `nhf_eq_of_hom_equiv`, `nhf_le_of_edge_injective`
Thm. 2.4, Table 1	`nhf_top_top_formula`, `nhf_top_succ_top`, `table_KnKm`
Thm. 2.5	`lower_bound`
Thm. 2.6	`nhf_top_triangleFree`
$\|C_n, K_3\|$, $\|C_n, K_2\|$	`nhf_cycle_K3`, `nhf_cycle_K2_even`, `nhf_cycle_K2_odd`
Thm. 2.7	`nhf_petersen_K2`
Thm. 2.8	`line_lower'`, `line_eq`, `line_cubic_pm`
Thm. 2.10	`sq_lower`
Thms. 2.11, 2.12	`hedetniemi_bound`, `nhf_tensor_cycles`
Thm. 2.13	`jaeger_lower`
Prop. 3.2, Lemma 3.3	`ceil_div_le_gamma`, `one_le_gamma`, `gamma_self`
Lemmas 3.4, 3.5, 3.7	`gamma_comp_le`, `nhf_le_gamma_mul`, `gamma_le_of_injective_hom`
Cor. 3.6, Prop. 3.8	`two_sided`, `gamma_top_top_eq_one`, `gamma_top_top_eq_top`
Lemma 3.9	`two_load_eq`, `two_mu_eq_deg`
Thm. 3.10	`two_mu_eq`, `gamma_K3_lower`, `gamma_K3_upper`
Thms. 3.11, 3.12	`gamma_cycle`, `gamma_knn`, `nhf_knn_K3`
Thm. 3.13	`gamma_petersen_K3`, `gamma_petersen_K5`
Thm. 3.14	`gamma_book`, `nhf_book_K2`, `gamma_friendship`, `nhf_friendship_K2`, `gamma_wheel`, `gamma_tri`
Thm. 3.15 and examples	`cycle_hom_iff`, `gamma_C6_C5`, `gamma_C15_C7`, ...
Prop. 4.1, Thm. 4.2	`msym_eq_zero_iff`, `no_triangle`, `msym_not_triangle`

Two parts of the paper are outside the formal development. Theorem 2.13 is formalized for arbitrary disjoint families of sets, each left by exactly one edge; that leaf blocks form such a family is not formalized. In Theorem 2.8 the equality is formalized given a balanced colouring, and in particular for cubic graphs with a perfect matching. The classical existence results of Remark 2.9 are not formalized.

6 Open Questions

1. **General targets.** Lemma 3.9 determines $\mu(f)$ for $f\colon G \to K_3$ from the degree sums of the colour classes. Is there an analogous identity for targets H other than K_3?

2. **When is the pigeonhole bound attained?** For cycles into K_3 it is attained exactly when $n \not\equiv 2 \pmod 3$ (Theorem 3.11). For d-regular G, Theorem 3.10 turns the question into one about realisable colour-class sizes. Characterise the pairs (G, H) for which $\gamma(G, H) = \lceil |E(G)|/|E(H)| \rceil$.

3. **Repairing m.** Does the edge-edit distance to a graph mapping to H, or m restricted to graphs with the same number of vertices, satisfy the triangle inequality?

4. **Petersen colourings.** Is $|L(G), L(P)|$ equal to the number of leaf blocks of the bridge-block tree for every cubic G? The case of bridgeless G is Jaeger's conjecture.

5. **Tightness of Theorem 2.11.** For which G, H is the bound tight for $K = K_2$ or $K = K_3$? Theorem 2.12 gives an infinite tight family for K_2. For large targets the bound fails to be tight, by [5].

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